AI Contract Hub

A Bilingual AI Assistant for Initial Review of Service Agreements

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Abstract

AI Contract Hub is a working prototype developed for the Digital Egypt Pioneers Initiative to support the initial review of Service Agreements. Through Telegram, users submit a PDF, DOCX file, or contractual text and receive a structured assessment generated through an n8n workflow and Google Gemini. The project combines document processing, bilingual response management, a twelve-rule legal checklist, and delivery controls. Its contribution is an accessible, repeatable review process that presents contractual evidence and suggested amendments for professional assessment. It supports legal judgment without replacing a qualified lawyer.

1. Problems and Objectives

Initial contract review requires lawyers to identify material obligations, assess their interaction, and explain risks in actionable terms. Repetitive document handling and inconsistent review structures can make this work difficult to organise, particularly across Arabic and English documents. The project aims to standardise the first review stage, identify material ambiguities and omissions, and produce traceable recommendations. Its scope includes complete Service Agreements and individual clauses, with conclusions limited to the text supplied.

2. System Design and Methodology

The workflow has three layers: input and document processing, AI analysis, and delivery with error handling. Telegram provides the user interface [1], while n8n coordinates routing and processing [2]. Input classification separates greetings and requests for instructions from contractual submissions. PDF processing extracts text; a dedicated DOCX route downloads the file, decompresses its archive, and reads the document XML. Unsupported formats receive an explanatory response. Extracted content is prepared for the Knowledge Agent, which uses Gemini to generate the review [3].

Language management detects Arabic or English from user messages, stores the preference against the conversation identifier, and retrieves it when a file arrives without explanatory text. A response-language resolver supplies one consistent preference to downstream steps. This separates the language of communication from the language of the uploaded contract. Processing notices indicate that work is underway, and long reports are divided for Telegram delivery.

3. Legal Review Framework

The twelve-rule checklist covers parties, scope of services, payment, term and renewal, termination, limitation of liability, indemnities, confidentiality, intellectual property, data protection, governing law, and dispute resolution. Instructions require contextual analysis, including interactions between clauses and existing safeguards. The current approach is checklist-guided prompting; it does not establish independent retrieval or verification of legislation. Missing clauses must be distinguished from provisions absent only because the user supplied an excerpt.

4. Review Output and Bilingual Functionality

Each finding is designed to identify the affected party, contractual reference, original excerpt, relevant checklist rule, practical consequence, recommendation, and proposed wording. Risks are classified as high, medium, or low in the response language. When an excerpt is in a different language, the report retains the original quotation and places a faithful translation beneath it. Suggested amendments and their placeholders follow the user's response language. The report also includes an executive summary, existing safeguards, negotiation priorities, and matters requiring human review.

5. Development Findings and Evaluation

Development runs have demonstrated report generation and Telegram delivery. Iterative testing exposed practical issues involving credential configuration, long messages, language preferences, and inconsistent output formatting. These informed revisions to workflow routing, message splitting, and review instructions. In particular, successful Arabic greetings did not ensure that later status messages and legal analysis used Arabic. This highlighted the need to carry one language preference throughout the workflow and to distinguish original evidence from generated commentary.

These observations establish functional progress, not validated legal accuracy or measured time savings. Formal evaluation should compare outputs against a lawyer-reviewed set of anonymised agreements and clauses. Test cases should cover both languages, PDF and DOCX inputs, uploads without captions, incomplete excerpts, long responses, and processing failures. Measures should include extraction completeness, material-issue detection, unsupported findings, citation accuracy, translation fidelity, language consistency, amendment quality, and successful delivery. No numerical performance claim is made at this stage.

6. Limitations and Responsible Use

Review quality depends on the available text, document structure, legal context, and model output. Scanned pages, complex tables, missing schedules, or extraction errors can affect conclusions. Checklist references do not prove compliance with applicable law, and generated quotations, risk ratings, and amendments require professional verification. Operational constraints include API availability, usage limits, and processing costs. Before use with confidential client material, the deployment requires documented access controls, retention settings, credential protection, and assessment of data handling across Telegram, n8n, and the model provider. Demonstrations should use public, synthetic, or appropriately anonymised documents.

7. Conclusion and Future Development

AI Contract Hub translates a defined legal review process into a practical bilingual workflow. It provides a foundation for organising initial findings and supporting lawyer-led review. Priorities for further development are documented benchmarking, stronger extraction checks, verified legal-source retrieval, and exportable Word or PDF reports. Expansion to additional agreement types should follow validation of the existing scope. The project demonstrates applied AI workflow development while retaining human accountability for legal conclusions.

References

[1] Telegram. Telegram Bot API. https://core.telegram.org/bots/api

[2] n8n. Documentation. https://docs.n8n.io/

[3] Google. Gemini API documentation. https://ai.google.dev/gemini-api/docs